

Project Proposal

Program description:

A two-player wordle game in which two players compete to see who can guess the word in the least amount of guesses. Upon entering the game, the player will be prompted to either 1) join an existing game or 2) create their own game. If they choose option 1, they will be prompted to input the join code. If they choose option 2, the program will output a join code for them to share with another player. Once both words have been given, players will be prompted to begin guessing. The end stats (winner, number of guesses per user) will be displayed once both players have finished guessing.

Communication protocols:

Protocol: Join / create game

Encoding: server → client (for create), client → server (for join)

Semantics: the game code as a single int (if player creates game, will receive a generated game code; if player joins game, will need to input an existing game code)

Error handling: output will be sent to user notifying them of success/failure of join/create

Protocol: Choose word

Encoding: client → server

Semantics: char * representing word choice. Server will read a string of exactly 5 from input (since words are all of length 5).

Error handling: We will error check the input length. We will also have a preset list of 5-letter words to choose from with which server will validate the user's guess.

Protocol: Guess word

Encoding: client → server

Semantics: see Choose word above

Error handling: will provide appropriate outputs to client if user inputs invalid word, and will output a string with information regarding the accurate of their guess (like traditional wordle)

Protocol: Game over

Encoding: server → client

Semantics: the game and user data represented as ints (but returned to both clients as a string). The server will store game information for each round between two users in a game struct and return the game over data to both users.

Error handling: will check game state and send appropriate messages to client.